

Practicum 6: Current-Condition Operational Hazards

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Question

What operational concerns are present in the latest saved weather observation?

Summary

The official ceiling-and-visibility weather category is **VFR**. The overall project screening result is **No listed hazard trigger**. No listed project hazard trigger is active.

Official weather categories and project-defined concern levels are separate. Neither result labels a flight safe or unsafe or grants aircraft-specific operating approval.

Data and method

The observation comes from saved data originally obtained from the [Aviation Weather Center Data API](#). Quarto rendering does not call the live API. The report validates its saved response and consumes the complete result from the project's central `operational_rules.py` module. It does not reproduce thresholds or concern aggregation.

- **Raw source file:** `tests/fixtures/metar-katl-success.json`
- **Processed source file:** Not applicable (assessment generated from committed fixture)
- **Station:** KATL (Atlanta/Hartsfield-Jackson Intl, GA, US)
- **AWC observation time:** 2026-07-29T19:52:00+00:00
- **AWC report time:** 2026-07-29T20:00:00.000Z
- **Retrieval time:** Not recorded for committed fixture
- **Assessment time:** 2026-07-29T20:00:00+00:00
- **Report render date:** 2026-07-29

Official flight category

Table 1: Official ceiling-and-visibility weather category

Field	Result
Category	VFR
Observed	Ceiling Ft Agl: 20000; Visibility Sm: 9.0; Awc Reported Category: VFR
Category trigger	Ceiling >3,000 ft AGL and visibility >5 SM.
Data complete	True
Operational context	Describes the point observation's official ceiling and visibility weather category without determining flight suitability.

Project-defined hazard screening

Table 2: Observed evidence and centralized project-defined concern results

Hazard	Observed	Concern	Confidence
Thunderstorm	Present Weather: Not reported	No listed hazard trigger	Standard
Convective cloud	Cloud Types: None reported	No listed hazard trigger	Standard
Freezing precipitation	Present Weather: Not reported; Automated Station Type: AO2	No listed hazard trigger	Standard
Wind	Sustained Kt: 9; Gust Kt: Not reported	No listed hazard trigger	Standard
Observation freshness	Observation Time: 2026-07-29T19:52:00+00:00; Age Minutes: 8.0	No listed hazard trigger	Standard

Thunderstorm

- **Result:** No listed hazard trigger
 - **Trigger applied:** No TS or VCTS token is present in decoded present weather.
 - **Operational context:** No listed thunderstorm trigger was found in this point observation.
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Convective cloud

- **Result:** No listed hazard trigger
 - **Trigger applied:** No structured cloud layer is typed CB or TCU.
 - **Operational context:** No listed convective-cloud trigger was found in the structured layers.
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Freezing precipitation

- **Result:** No listed hazard trigger
 - **Trigger applied:** No FZRA or FZDZ token is present in decoded present weather.
 - **Operational context:** No listed freezing-precipitation trigger was found in this point observation.
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Wind

- **Result:** No listed hazard trigger
 - **Trigger applied:** Sustained wind and any reported gust are <25 kt.
 - **Operational context:** Compare reported wind with runway-relative conditions and applicable aircraft, operator, and personal limits; these project thresholds are not universal limits.
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Observation freshness

- **Result:** No listed hazard trigger
- **Trigger applied:** Observation age is <=75 minutes and no more than 5 minutes in the future.
- **Operational context:** No project-defined freshness trigger was reached.

Source basis

- [FAA AIM categorical outlook definitions](#): Defines VFR, MVFR, IFR, and LIFR ceiling and visibility categories; these weather categories are not aircraft operating approvals.

- [FAA AIM thunderstorm avoidance and METAR weather codes](#): Defines TS and VC coding and describes thunderstorm hazards.
- [FAA AIM icing definitions](#): Defines freezing rain and freezing drizzle as freezing precipitation.
- [FAA AIM automated weather observation capabilities](#): AO1 identifies a station without a precipitation discriminator; AO2 identifies a station with one.
- [FAA AIM aviation weather product wind criteria](#): Provides contextual anchors for categorical-outlook WIND notation, sustained surface wind AIRMETs, and severe-thunderstorm gust criteria.
- [FAA Personal Minimums for Wind](#): Explains that wind limits must reflect the pilot, aircraft, runway, and operating context.
- [NWS METAR overview](#): Describes the routine hourly METAR cadence and one-hour validity.
- [NWS aviation data retrieval guidance](#): Describes expected central processing time for hourly observations.

Interpretation

The centralized assessment reports the official weather category as **VFR** and the highest active project-defined concern as **No listed hazard trigger**. Data completeness is **True**. Informational screening only. Official flight categories are weather classifications, and project-defined concern levels are not universal aircraft operating limits or official flight guidance. Review applicable weather products, procedures, limitations, and pilot or dispatcher judgment.

Rebuild

After installing the locked dependencies with `uv sync`, rebuild the PDF from the repository root with:

```
uv run quarto render reports/practicum-6.qmd --to pdf --output-dir ../output/pdf
```